

Video Analysis Progress Tracking - Deployment Fix Summary

Issue Identified

The “Premium video analysis with progress tracking” version (commit `ec8277f`) failed to deploy because:

Root Cause: The production database does not have the new tables required by the feature:

- `VideoAnalysisProgress` - Tracks user progress per skill type
- `VideoAnalysisAchievement` - Defines available achievements
- `VideoAnalysisUserAchievement` - Links users to unlocked achievements

Verification Completed

Build Status

-  Local build succeeds without errors
-  All TypeScript type checks pass
-  All dependencies properly installed
-  No missing imports or broken references

Code Quality

-  Prisma schema includes all 3 new models
-  API endpoints properly configured:
 - `/api/video-analysis/progress` - Returns user progress data
 - `/api/video-analysis/achievements` - Returns achievements with unlock status
-  UI components updated to display progress and achievements
-  Seed script created with 18 achievements

Database Schema

```

VideoAnalysisProgress (New)
├── id: String (PK)
├── userId: String (FK → User)
├── skillType: String (unique with userId)
├── averageScore: Float
├── totalShots: Int
├── bestScore: Float
├── sessionsCount: Int
├── lastAnalyzed: DateTime
└── improvementRate: Float

VideoAnalysisAchievement (New)
├── id: String (PK)
├── code: String (unique)
├── name: String
├── description: String
├── icon: String
├── category: String
├── threshold: Float
├── thresholdType: String
└── points: Int

VideoAnalysisUserAchievement (New)
├── id: String (PK)
├── userId: String (FK → User)
├── achievementId: String (FK → VideoAnalysisAchievement)
├── unlockedAt: DateTime
└── value: Float

```



Files Created

1. Deployment Guide

File: VIDEO_ANALYSIS_DEPLOYMENT_GUIDE.md

- Comprehensive step-by-step deployment instructions
- Database migration options (Prisma Migrate vs DB Push)
- Environment variable checklist
- Troubleshooting section
- Success criteria

2. SQL Migration Script

File: manual-migration-video-analysis.sql

- Manual SQL migration for direct database application
- CREATE TABLE statements for all 3 new tables
- Indexes and constraints
- Verification queries

3. Verification Script

File: scripts/verify-video-analysis-setup.ts

- Automated verification of database setup
- Checks all tables exist
- Verifies 18 achievements are seeded

- Tests relationships and indexes
- Returns exit code 0 on success, 1 on failure

4. Package.json Scripts

Added convenient npm commands:

```
{
  "seed:video-achievements": "tsx scripts/seed-video-achievements.ts",
  "verify:video-analysis": "tsx scripts/verify-video-analysis-setup.ts"
}
```

Deployment Checklist

Pre-Deployment (Required Actions)

- ☐ **Apply Database Migration** (Choose one option):

Option A: Prisma DB Push (Recommended for quick deployment)

```
bash
cd /home/ubuntu/mindful_champion/nextjs_space
DATABASE_URL="your_production_url" npx prisma db push
```

Option B: Manual SQL (If you prefer direct SQL)

```
bash
psql $DATABASE_URL < manual-migration-video-analysis.sql
```

- ☐ **Seed Achievements**

```
bash
cd /home/ubuntu/mindful_champion/nextjs_space
npm run seed:video-achievements
# Or with custom database:
# DATABASE_URL="..." npm run seed:video-achievements
```

- ☐ **Verify Setup**

```
bash
npm run verify:video-analysis
```

- ☐ **Check Vercel Environment Variables**

- DATABASE_URL - Production PostgreSQL connection
- NEXTAUTH_SECRET - Session secret
- NEXTAUTH_URL - https://mindfulchampion.com
- RESEND_API_KEY - Email service
- AWS_* - Video upload credentials (S3)

Post-Deployment (Verification)

- ☐ **Build succeeds** in Vercel dashboard
- ☐ **Health check:** `curl https://mindfulchampion.com/api/video-analysis/progress`
- ☐ **Test UI:** Upload a video and check for progress tracking
- ☐ **Achievement unlock:** Verify first analysis unlocks “First Steps” achievement

Quick Deployment Commands

```
# 1. Navigate to project
cd /home/ubuntu/mindful_champion/nextjs_space

# 2. Apply database changes (production)
DATABASE_URL="your_prod_url" npx prisma db push

# 3. Seed achievements (production)
DATABASE_URL="your_prod_url" npm run seed:video-achievements

# 4. Verify setup (production)
DATABASE_URL="your_prod_url" npm run verify:video-analysis

# 5. Commit and push (triggers Vercel deployment)
git add -A
git commit -m "Video analysis progress tracking - deployment ready"
git push origin master
```



Expected Results

After Successful Deployment

1. Database Tables:

- 3 new tables created
- All indexes and foreign keys in place
- 18 achievements seeded

2. API Endpoints:

- `/api/video-analysis/progress` returns user stats
- `/api/video-analysis/achievements` returns all achievements

3. User Experience:

- Progress tracking dashboard shows improvement over time
- Skill-specific metrics for each shot type
- Achievement notifications on unlock
- Historical charts with session data

4. Achievement Categories:

- **Milestones:** 4 achievements (First Steps → Analysis Pro)
- **Speed:** 3 achievements (30 MPH → 50 MPH)
- **Score:** 3 achievements (80+ → 95+)
- **Accuracy:** 3 achievements (Perfect Shot → Flawless)
- **Improvement:** 3 achievements (10% → 50%)
- **Consistency:** 2 achievements (Weekly → Monthly)



Important Notes

1. Database Migration is Mandatory

The deployment will fail without applying the database migration first. The new API endpoints and UI components expect these tables to exist.

2. Seed Script Should Run Once

Run the achievement seed script only once. It uses `upsert` so it's safe to run multiple times, but unnecessary.

3. Backward Compatibility

The feature is backward compatible. Users without video analyses won't see errors - the UI gracefully handles empty states.

4. Performance Considerations

- Indexes are added on `userId` fields for fast queries
- Unique constraints prevent duplicate progress records
- Cascading deletes ensure data integrity



Troubleshooting

“Table does not exist” Error

Solution: Run the database migration:

```
DATABASE_URL="prod_url" npx prisma db push
```

“No achievements found”

Solution: Run the seed script:

```
npm run seed:video-achievements
```

Build succeeds but API returns errors

Solution: Check Vercel logs and verify `DATABASE_URL` is set in environment variables.

Verification script fails

Solution: Check the specific error message. Common issues:

- `DATABASE_URL` not set or incorrect
- Tables not created (run migration)
- Achievements not seeded (run seed script)



Support Resources

1. **Deployment Guide:** `VIDEO_ANALYSIS_DEPLOYMENT_GUIDE.md`
2. **SQL Migration:** `manual-migration-video-analysis.sql`
3. **Verification Script:** `scripts/verify-video-analysis-setup.ts`
4. **Seed Script:** `scripts/seed-video-achievements.ts`



Feature Highlights

For Users

- Track improvement over time with visual charts
- Skill-specific progress for each shot type

- 🏆 18 achievements across 6 categories
- 📊 Historical analysis comparison
- 💡 Quantified improvement percentages

For Developers

- 🔒 Type-safe Prisma models
- 🚀 Optimized database queries with indexes
- ✅ Comprehensive verification tooling
- 📖 Detailed documentation
- 🛠️ Easy local testing setup

Summary

Status: ✅ Code is production-ready and verified

Blocker: Database migration needs to be applied to production

Action Required: Run migration → Seed achievements → Deploy

Estimated Time: 5-10 minutes

All code changes are complete and tested. The only remaining step is applying the database schema to production and triggering the deployment.

Last Updated: December 23, 2025

Verified By: DeepAgent

Build Status: ✅ Passing

Type Checks: ✅ Passing

Ready for Deployment: ✅ Yes (after DB migration)